

TCP Round-Trip Time Estimation

and Retransmission Timeout

1. Introduction

TCP, like our `rdt` protocol, uses a **timeout/retransmit mechanism** to recover from lost segments. The selection of the timeout interval is one of the most subtle issues in protocol design.

- The timeout must be **larger than the round-trip time (RTT)**.
- It should not be **too much larger** to avoid excessive recovery delays.

2. Estimating the Round-Trip Time

2.1 SampleRTT

The **SampleRTT** for a segment is the time between when the segment is passed to IP and when its acknowledgment is received.

Measurement Principles

- Measure only **one segment at a time** (roughly once per RTT).
- **Karn's Algorithm:** Never compute **SampleRTT** for retransmitted segments.

2.2 EstimatedRTT (Smoothing)

Because **SampleRTT** values fluctuate, TCP maintains an average called **EstimatedRTT** using an **Exponential Weighted Moving Average (EWMA)**.

EstimatedRTT Formula

$$\text{EstimatedRTT} = (1 - \alpha) \cdot \text{EstimatedRTT} + \alpha \cdot \text{SampleRTT}$$

Recommended value: $\alpha = 0.125$ (1/8)

2.3 DevRTT (Variation)

DevRTT estimates how much **SampleRTT** typically deviates from **EstimatedRTT**.

DevRTT Formula

$$\text{DevRTT} = (1 - \beta) \cdot \text{DevRTT} + \beta \cdot |\text{SampleRTT} - \text{EstimatedRTT}|$$

Recommended value: $\beta = 0.25$

3. Setting the Timeout Interval

TCP sets the timeout equal to the `EstimatedRTT` plus a "safety margin" that accounts for variability.

TimeoutInterval Formula

$$\text{TimeoutInterval} = \text{EstimatedRTT} + 4 \cdot \text{DevRTT}$$

3.1 Dynamic Management Rules

- **Initial Value:** Set to **1 second**.
- **Exponential Backoff:** On a timeout, the `TimeoutInterval` is **doubled** to avoid further premature timeouts during congestion.
- **Recovery:** When an acknowledgment arrives for a non-retransmitted segment, `TimeoutInterval` is recalculated using the formula above.